

Development of hybrid solar-assisted heat pump dryer for drying paddy

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ABSTRACT

This recent study evaluated a hybrid solar-assisted heat pump dryer (HS-AHPD) for drying paddy and its product quality. R22 is used as a working fluid in the heat pump system during the drying experiment. The HS-AHPD decreased the paddy's weight from 420 kg (31.67% db) to 370.60 kg (16.18% db) in 5.5 h with an average air temperature of 62.9 °C and an average relative humidity of 16.1% at a mass flow rate of 0.195 kg/s. The average drying rate, specific moisture extraction rate (SMER), and specific energy consumption (SEC) were 8.34 kg/s, 0.44 kg/kWh, and 4.69 kWh/kg, respectively. The HS-AHPD's average thermal efficiency and the exergy efficiency of the drying section were found up to 29.1% and 18.4%, respectively. The furnace and the solar collector contributed 19.7% and 12.9% of energy to the HS-AHPD, respectively. The improvement potential (IP) was in the range of 628.8–824.4W. The drying resulted in higher-quality products due to higher head rice yield ($93.10 \pm 1.044\%$), lower broken rice ($4.41 \pm 0.737\%$), and lower rice grouts ($1.11 \pm 0.271\%$).

Nomenclature

Asc	solar collector area (m ²)
BR	broken rice (%)
C _a	specific heat of air (Jkg ⁻¹ °C ⁻¹)
C _{da}	specific heat of dry air (Jkg ⁻¹ °C ⁻¹)
COP	coefficient of performance
CV	caloric value (kcal/kg)
EC _{BF}	energy contribution from biomass furnace (%)
EC _{SC}	energy contribution from solar collector (%)
Ex _{dsi}	exergy input (J/s)
Ex _{dso}	exergy output (J/s)
FBD	fluidized bed dryer
GD	greenhouse dryer

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GG	percentage of grain grouts (%)
H_{fg}	latent heat (water vaporization) (kJ/kg)
HRV	head rice yield (%)
I	current (A)
I_T	solar radiation (W/m^2)
m	mass (kg)
$\dot{m}_a$	air mass flow rate (kg/s)
$\dot{m}_{bf}$	biomass fuel consumption rate (kg/s)
MC	moisture content (%)
$\dot{m}_{da}$	dry air mass flow rate ($kg_{dry\ air}/s$)
$\dot{m}_{water}$	drying rate (kg/min)
PP	payback period (year)
Q_{EP}	thermal energy produced (W)
Q_I	thermal energy input (W)
T	temperature ($^{\circ}C$)
t	time (min)
V	voltage (V)
W_T	weight (g)
$\cos \varphi$	power factor
η_{BF}	efficiency of the biomass furnace (%)
η_{Ex}	exergy efficiency (%)
η_{SC}	efficiency of the solar collector (%)
η_{th}	efficiency of the thermal dryer (%)

Subscripts

amb	ambient
Bbf	biomass furnace blower
Bd	dryer blower
bf	biomass furnace
bmf	biomass fuel
C	compressor
chp	condenser of heat pump
db	dry basis
ds	drying section
HP	heat pump
i	inlet
Mbe	bucket elevator motor
Mdr	discharge roller motor
Mvf	vibratory feeder motor
o	outlet
sc	solar collector
THR	total head rice
w	wet bulb
wb	wet basis

1. Introduction

Drying is traditionally used to preserve agricultural products and primarily to lower the moisture content of agricultural products to a particular degree to extend their storage time [1]. Some developing and tropical countries have two popular drying methods to dry agricultural products: the open sun and the solar drying methods. The former is simple and low-cost but requires a long time and results in low product quality, while the latter requires a shorter time but produces a high-quality product and abundant solar radiation. However, the drying operation is inapplicable on cloudy and rainy weathers [2]. Additionally, drying is an energy-intensive process that requires a large energy input because of the high latent heat of water evaporation; approximately 12% of the total energy used in all industries in developed countries is consumed by drying processes [3,4].

In order to overcome the limitations of the open sun and the solar drying methods or the current drying techniques, and also to reduce the energy consumption, it is necessary to select an efficient heating system. Combination of a heat pump system with a solar dryer provides a solution to the current drying techniques. Heat pump systems are energy efficient devices due to its ability to deliver more amount of heat energy than the work input it takes, and are also environmentally friendly technologies [5,6]. The combination of a heat pump system with a solar dryer is known as a solar-assisted heat pump dryer (SAHPD). In general, the SAHPD consists of two

Table 1
SAHPD using various type of drying chambers.

No.	Drying chamber type	Products	MC (%wb)	Temp. (°C)	Time (h)	Energy Contribution (%)		SMER (kg/kWh)	Efficiency (%)				COP _{HP}	PP (y)	Refs.
			M _{ci}												
			M _{cf}			SC	BF		η _{th}	η _{EX}	η _{SC}	η _{BF}			
1	Tray	Mushroom	13.24 db 0.07 db	45–55	3.83–3.17	-	-	0.26–0.92	-	-	-	-	-	-	[19]
2	Tray	Copra	52 9.2	43.2	40	-	-	0.79	-	-	-	-	-	-	[20]
3	Tray	Cassava chips	61 10.5	45	9	44.6	-	0.47	30.9	-	55	-	3.38	-	[2]
4	Fluidized bed	Mint leaves	5.67 db 0.17 db	50	11	-	-	-	50	26	-	-	-	-	[21]
5	Tray	Banana	74 19	54.8	21	-	-	0.6	-	-	-	-	2.72	-	[22]
6	Tray	Chili peppers	80–85 10–11	53	24	-	-	-	33.2	-	-	-	3.17	1.9	[23]
7	Tray	Banana chips	83.5 11.5	65.1	3.75	-	-	1.45	-	45.08	-	-	2.04	46 months	[1]
8	Bine	corn	12.9 12.5	-	42	-	-	1.93	-	40.27	63	-	5.03	0.33	[24]
9	Tray (Integrated with biomass furnace)	Red chili	4.26 db 0.08 db	68.4–71.8	11	14.7	37.9	0.14	9.03	-	35.1	30.7	3.84	-	[25]
10	Fluidized bed (Integrated with biomass furnace)	Rice	27.7 14	78.9–81.6	0.38	11	30	0.24	15.4	41.3	50.5	77.5	3.7	1.6	[26]

COP_{HP}: coefficient of performance of heat pump; BF: biomass furnace; MC: moisture content; M_{ci}: initial moisture content; M_{cf}: final moisture content; PP: payback period of drying system; SC: solar collector; SMER: specific moisture extraction rate; η_{BF}: efficiency of biomass furnace; η_{EX}: exergy efficiency for drying chamber; η_{SC}: efficiency of solar collector; η_{th}: thermal dryer efficiency.

circuits, namely the drying air circuit and the refrigerant circuit. The drying air circuit consists of a solar collector, drying chamber, and blower. Meanwhile, the refrigerant circuit (heat pump circuit) consists of a compressor, condenser, expansion valve or capillary tube, and evaporator. The evaporator of the heat pump is used to dehumidify the process air leaving the drying unit, which is a particularly attractive feature for drying, while the condenser of heat pump is used to heat the drying air by using the heat released by the condensing refrigerant.

The performance of the SAHPD is affected by the performance of the heat pump, while the performance of the heat pump is affected by the ability of the evaporator to remove moisture from the air and the ability of the condenser to heat the drying air [7,8].

The SAHPD offer some advantages: the drying process can run continuously, needs a shorter time, consumes low energy, and produces good product quality [3].

Several studies said that SAHPDs have been used to dry agricultural or biologically active products, including rice [9,10], green beans [11], red chili [12], agricultural food grains [13], tomato, strawberry, mint, and parsley [14], radish, pepper, and mushroom [15], mango [16], wheat [17], and tibetan medicinal materials [18].

Sevik et al. [19] studied an SAHPD's performance by using a tray-type drying chamber for mushrooms. They found the coefficient of performance (COP) of the system and SMER were 2.1–3.1 kg/kWh and 0.26–0.92 kg/kWh, respectively. The dryer decreased the mushrooms' moisture content (MC) from 13.24 db to 0.07 db with air temperatures of 45 °C and 55 °C and a mass flow rate of 310 kg/h within 230–190 min.

Mohanraj [20] investigated an SAHPD's performance using a drying chamber similar to that used by Sevik et al. [13] for copra in a hot-humid condition. He reported that the dryer's COP was 2.31–2.77 with an average value of 2.54, while the average SMER was found up to 0.79 kg/kWh. The dryer reduced copra's MC from 52% wb to 9.2% wb within 40 h at an average temperature of 43.2 °C.

Yahya et al. [2] did the same but for cassava chips, finding that the average SMER, dryer thermal efficiency, and pickup efficiency were 0.47 kg/kWh, 30.9%, and 43.6%, respectively. The heat pump's COP and the solar collector's efficiency were up to 3.38 and 55%, respectively, while the solar collector contributed approximately 44.6% of energy to the SAHPD. The SAHPD reduced cassava's MC from 61% to 10.5% wb within 9 h with a mass flow rate and an average temperature of 0.124 kg/s and 45 °C, respectively.

Ceylan and Gurel [21] examined experimentally an SAHP-FBD for drying mint leaves and revealed that the time needed in the SAHP-FBD to decrease the leaves' MC from 5.67 db to 0.17 db with average air temperature and velocity of 50 °C and 3.8 m/s, respectively, was approximately 11 h. The dryer's average exergy efficiency and energy were up to 26% and 50%, respectively.

Kuan et al. [22] evaluated numerically an SAHPD's performance using the identical type drying chamber for banana and found that the heat pumps' average COP and SMER were 2.72 and 0.6 kg/kWh, respectively. The banana's MC decreased from 74% wb to 19% wb within 21 h at 54.8 °C.

Naemsai et al. [23] observed an SAHP-GD's performance using the same type of drying chamber for chili peppers and reported that the SEC, drying efficiency, and COP were 2.21 kWh/kg, 33.2%, and 3.17, respectively. The dryer reduced the chili peppers' MC from 80 to 85% wb to 10–11% wb within 24 h at 53 °C and an average relative humidity of 41%. The dryer's payback period was 1.9 years.

Singh et al. [1] examined experimentally an SAHPD using a tray-type drying chamber for banana chips and reported that the time needed in the dryer to reduce the banana chips's MC from 83.5% wb to 11.5% wb with an average drying temperature of 65.1 °C and an air velocity of 1 m/s was approximately 225 min. The heat pump's COP, SMER, and the exergy efficiency were 2.04, 1.45 kg/kWh and 45.08%, respectively. The system's payback period was 46 months.

Gu et al. [24] observed experimentally a novel solar-assisted heat pump bin dryer for corn and reported that the solar collector's efficiency and the pump's COP were 63% and 5.03, respectively. The dryer decreased the corn's MC from 12.9% wb to 12.5% wb within 42 h. The SMER was 1.934 kg/kWh, while and the exergy efficiency was 40.27%. The drying system's payback period was 0.33 years.

However, SAHPD has a drawback: low drying air temperature when the day is cloudy and rainy can make the drying operation incomplete due to the heat pump's limited condenser to produce heat energy and the solar collector's limited ability to collect the sun energy. To overcome these problems, an SAHPD can be integrated with a biomass furnace, which uses biomass as a heat energy source.

Yahya [25] investigated the performance of an SAHPD integrated with biomass furnace using a tray-type drying chamber for red chili. He reported that the average SMER was 0.14 kg/kWh and the dryer thermal efficiency was 9.03%. Furthermore, the heat pump's COP, the solar collector's efficiency, and the biomass furnace' efficiency were up to 3.84, 35.1%, and 30.7%, respectively. Meanwhile, the furnace and the solar collector contributed approximately 37.8% and 14.7% of energy, respectively, to the dryer. The red chili's MC was reduced from 4.26 db to 0.08 db at a mass flow rate of 0.124 kg/s in 11 h. The air temperature was in the range of 68.4–71.8 °C by means of the biomass furnace.

Yahya et al. [26] observed an SAHP-FBD integrated with biomass furnace for rice and found that the average SMER, SEC, dryer thermal efficiency, pickup efficiency, and exergy efficiency were 0.24 kg/kWh, 4.76 kWh/kg, 15.4%, 43.8%, and 41.3%, respectively. They also found that the heat pump's COP, the solar collector's efficiency, and the biomass furnace's efficiency were up to 3.7, 50.5%, and 77.5%, respectively. Furthermore, the furnace contributed approximately 30% of energy, while and the solar collector did so about 11% to the dryer. From 27.7% wb, the rice's MC reduced to 14% wb within 22.95 min at a mass flow rate of 0.1037 kg/s. Yahya et al. set the air temperature ranging from 78.9 to 81.6 °C using the furnace. The dryer's payback period was 1.6 years. Table 1 shows the results.

Indonesia can produce 54.42 million tons of paddy and 236 million tons of biomass, annually [27,28]. The daily average solar radiation it gets reaches 4 kWh/m² [29]. Paddy, after being harvested, generally contains high moisture, ranging from 20% to 27% (wet basis) [30]. For this reason, for long-term storage and milling, it should be dried to have a MC of 14% wb [31]. Solar energy and biomass are useable as sources of heat energy for desiccating it.

We found no research assessing HS-AHPD for drying paddy. This study, therefore, was to fabricate and assess the performance of a

solar-assisted heat pump recirculating mixed-flow dryer integrated with a biomass furnace for drying paddy and the quality of drying products.

2. Material and method

2.1. Experimental set-up

Initially, we designed an HS-AHPD for drying paddy and then installed it. The dryer comprised several main components, namely a drying column, a biomass furnace, a heat pump system, a solar collector, a bucket elevator, a vibratory feeder, and a blower, as presented in Fig. 1. The drying column was prepared to have a storage section, a drying section, and a discharge section. The drying section had inlet and outlet air ducts arranged horizontally, and it was wedge-shaped. The discharge section comprised a support structure, a roller, a chain, a motor, a gear box, and a sprocket, while the biomass furnace consisted of a combustion chamber, a blower, a heat exchanger pipe, and a chimney. The heat pump system was built up from a capillary tube, an evaporator, a compressor, and a condenser. The solar collector used a single-pass flat plate type and consisted of a case, a transparent cover glass, an absorber finned plate, and an insulator. Two solar collectors were used and connected in series. The bucket elevator consisted of a support structure, a casing, a motor, a pulley, buckets, an elevator belt, a gear box, a feed hopper, and a discharge sprout. The vibratory feeder consisted of a support structure, a trough, a motor, a v-belt, a pulley, and a bearing. The dimensions of the drying column, the air duct of the drying section, and the biomass furnace were identical to those for a pilot-scale biomass-assisted recirculating mixed flow dryer previously presented by Yahya et al. [32]. Yahya [33] described the solar collector and the heat pump dimensions, as shown in the previous solar-assisted heat pump dryer. The components specification of the heat pump system can be seen in Table 2.

2.2. Experimental procedure

The researchers carried out the drying experiment from 9:30 a.m. to 3:30 p.m. for evaluating the performance of the HS-AHPD with a capacity of approximately 420 kg. The fresh paddy cosikan cultivar was bought from a local farmer. During the experiment, the air and paddy temperatures in the different positions of the drying system were measured using a thermocouple. The solar radiation and air flow rate were observed using a pyranometer and an anemometer, respectively. A data logger was used to record the solar radiation, air temperature, and paddy temperature. A grain moisture tester was employed to measure the paddy's moisture content. The paddy and the biomass fuel were weighed using a weighing scale, and the biomass fuel used was coconut-shell charcoal. The solar radiation, air temperature, paddy temperature, and paddy moisture content were recorded every 0.5 h. The following equation was employed to estimate the uncertainties [34]:

$$W_R = \left[\left(\frac{\partial R}{\partial x_1} w_1 \right)^2 + \left(\frac{\partial R}{\partial x_2} w_2 \right)^2 + \dots + \left(\frac{\partial R}{\partial x_n} w_n \right)^2 \right]^{1/2} \quad (1)$$

2.3. Analyses of performance and quality

The drying characteristics, the HS-AHPD performance, the dryer components performance, and the drying product quality were estimated using the equations in Table 3.

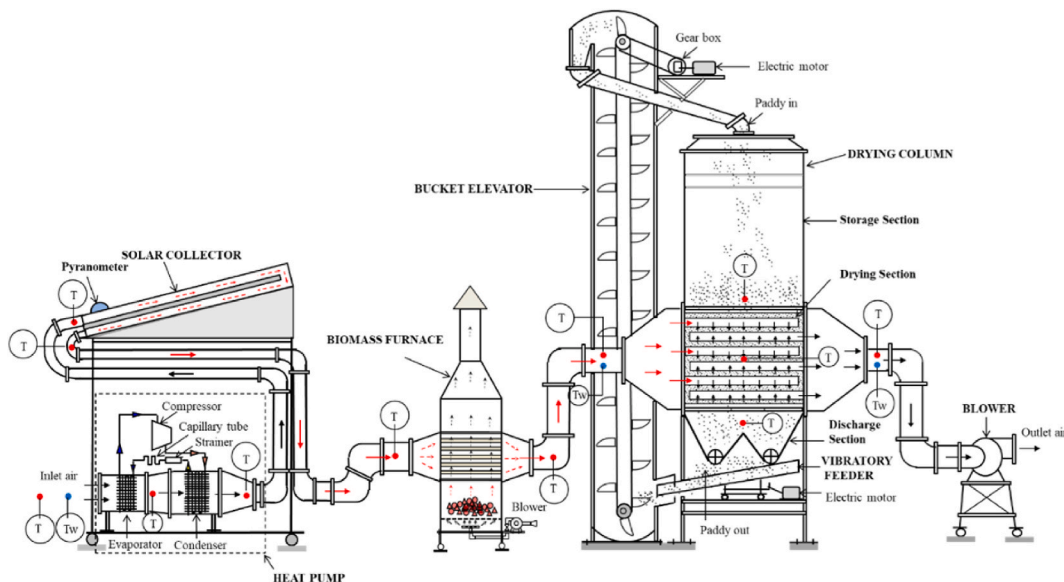


Fig. 1. Schematic of the HS-AHPD.

Table 2
Components specification of heat pump system.

1	Evaporator a. Type b. Material c. Tube diameter	Cross flow fin and tube Aluminum 9.52 mm
2	Condenser a. Type b. Material c. Tube diameter	Cross flow fin and tube Aluminum 9.52 mm
3	Compressor a. Type b. Power input c. Circular velocity	Hermetic 1HP 2800 rpm
4	Capillary tube a. Material b. Tube diameter	Copper 2 mm
5	Working fluid a. Refrigerant	R22

3. Results and discussion

3.1. Performance evaluation

Table 4 shows instruments' characteristics in this experiment and uncertainties in the drying performance. The relationship of the weight and the MC of the paddy to the drying time is illustrated in Fig. 2. In the HS-AHPD, the time needed for reducing the paddy's weight from 420 kg with MC of 31.7% db (24.1% wb) to 370.6 kg with MC of 16.2% db (13.9% wb) was approximately 5.5 h with an average air temperature of 62.9 °C and an average relative humidity of 16.1%, at a mass flow rate of 0.195 kg/s. As seen in Fig. 2, the paddy's weight and MC decreased with prolonging drying time. The decrease was faster in the drying experiment's initial period, followed by the rate decrease.

Fig. 3 depicts the variation of the DR and the SMER with the drying time. The DR and SMER ranged from 1.29 kg/s to 27.29 kg/s and from 0.71 kg/kWh to 1.47 kg/kWh, with average values of 8.34 kg/s and 0.44 kg/kWh, respectively. In Fig. 3, the SMER was significantly dependent on the DR; a higher DR corresponded to a higher SMER because of the great difference in partial vapor pressure between the paddy and the air drying that led to a high acceleration of moisture migration.

Fig. 4 presents the relationship of the DR to the paddy's MC. As shown in Fig. 4, the DR significantly depended on the paddy's MC; i. e., a higher MC was related to a higher DR. It was also shown that, during the drying, the constant rate was ignored, and the drying only happened in the falling rate period. It occurred since the DR depended on the nature of MC movement from the material inner part to the surface. From the curve of the DR with the paddy's MC, the regression equations were created as $DR = -33.37 + 2.1286 (MC)$ and $R = 0.9629$.

Fig. 5 indicates the variations of the SEC versus the drying time. The SEC varied from 0.68 kWh/kg to 14.08 kWh/kg, with an average of 4.69 kWh/kg. Fig. 5 also presents the variations of the STEC and SEEC versus the drying time. The former was in the range of 0.51–10.48 kWh/kg, with an average of 3.53 kWh/kg, while the latter was in the range of 0.17–3.59 kWh/kg, with an average of 1.17 kWh/kg. As shown in Fig. 5, the three parameters rised with increasing drying time due to the lowering DR with increasing drying time.

Fig. 6 presents the graph of the thermal efficiency of the HS-AHPD and the exergy efficiency of the drying section versus the drying time. The thermal efficiency of the HS-AHPD ranged from 4.7% to 96.0%, with an average of 29.1%, while the exergy efficiency of the drying section ranged from 14.4 to 25.3%, with an average of 18.4%. Fig. 6 also presents the graph of the IP versus the drying time. The IP ranged from 628.8 to 824.4 W, with an average of 735.62 W. As seen in Fig. 6, the HS-AHPD's thermal efficiency was high at the beginning of the drying and lowering at the end. The high thermal efficiency at the beginning of the drying was because the DR was very high and vice versa.

Fig. 7 shows the graph of the energy contributed by the biomass furnace and the solar collector versus the drying time. Both components contributed energy to the HS-AHPD in the ranges of 17.3–22.4% and 6.6–12.9%, respectively, with average values of 19.7% and 12.9%, respectively.

Fig. 8 presents the variation of the efficiency of the solar collector, the biomass furnace and the heat pump's COP versus the drying time. The solar collector's and the furnace's efficiencies were in the ranges of 55.4–81.0% and 27.9–39.0%, respectively, with average values of 72.2% and 33.03%, respectively, while the heat pump's COP was in the range of 2.81–3.26, with an average value of 3.06.

3.2. Evaluation of the quality of dried products

In the drying process, the product quality must be considered because it greatly affects the market value. It can be expressed in milling qualities such as the percentages of head rice yield, rice grouts, and broken rice. By milling the paddy during the desiccation process in the HS-AHPD, they were $93.10 \pm 1.044\%$, $4.41 \pm 0.737\%$, and $1.11 \pm 0.271\%$, respectively. The Indonesian national rice quality standard (SNI No.01-6128-2015) [44] reveals that the rice quality produced by the HS-AHPD at air temperatures of 60.0–64.8 °C belongs to in the premium class. The rice quality is high because the head rice yield is high and broken rice and rice groats are low, in percentages.

Table 3
Analyses of drying characteristics, performance, and quality.

Parameter	Formula Eq.no.	Ref.
<i>Drying characteristics</i>		
Moisture content of paddy (dry basis)	$M_{dbpd} = \frac{m_{wetpd} - m_{dpd}}{m_{dpd}} \quad (2)$	[35]
Paddy moisture content (wet basis)	$M_{wbpd} = \frac{m_{wetpd} - m_{dpd}}{m_{wetpd}} \quad (3)$	[35]
Drying rate (DR)	$\dot{m}_{water} = \frac{M_{dbpd,t+dt} - M_{dbpd,t}}{dt} \quad (4)$	[15,28]
<i>Drying system performance</i>		
Specific energy consumption	$SEC = \frac{Q_{isc} + Q_{lbmf} + W_C + W_{Bd} + W_{Bbf} + W_{Mt}}{\dot{m}_{water}} \quad (5)$	[36,37]
Total electrical energy input of electrical motor	$W_{Mt} = W_{Mbc} + W_{Mvf} + W_{Mdr} \quad (6)$	[36,37]
Specific electrical energy consumption	$SEEC = \frac{W_C + W_{Bd} + W_{Bbf} + W_{Mt}}{\dot{m}_{water}} \quad (7)$	[25,26]
Specific thermal energy consumption	$STEC = \frac{Q_{isc} + Q_{lbmf} + W_C}{\dot{m}_{water}} \quad (8)$	[25,26]
Energy contribution from biomass furnace	$EC_{BF} = \frac{Q_{Ebf}}{Q_{isc} + Q_{lbmf} + W_C + W_{Bd} + W_{Bbf} + W_{Mt}} \times 100\% \quad (9)$	[35]
Energy contribution from solar collector	$EC_{sc} = \frac{Q_{Esc}}{Q_{isc} + Q_{lbmf} + W_C + W_{Bd} + W_{Bbf} + W_{Mt}} \times 100\% \quad (10)$	- [15,26,36]
Specific moisture extraction rate	$SMER = \frac{\dot{m}_{water}}{Q_{isc} + Q_{lbmf} + W_C + W_{Bd} + W_{Bbf} + W_{Mt}} \quad (11)$	[19,37]
Thermal dryer efficiency	$\eta_{th} = \frac{\dot{m}_{water} H_{fg}}{Q_{isc} + Q_{lbmf} + W_C + W_{Bd} + W_{Bbf} + W_{Mt}} \quad (12)$	[14,25]
Exergy efficiency of drying section	$\eta_{Ex} = \frac{Ex_{dso}}{Ex_{dsi}} = 1 - \frac{Ex_{loss}}{Ex_{dsi}} \quad (13)$	[35,38,39]
Exergy losses	$Ex_{loss} = Ex_{dsi} - Ex_{dso} \quad (14)$	[35,38,39]
Exergy inflow for drying section	$Ex_{dsi} = \dot{m}_{da} C_{Pda} \left[(T_{i,ds} - T_{amb}) - T_{amb} \ln \frac{T_{i,ds}}{T_{amb}} \right] \quad (15)$	[35,38,39]
Exergy outflow for drying section	$Ex_{dso} = \dot{m}_{da} C_{Pda} \left[(T_{o,ds} - T_{amb}) - T_{amb} \ln \frac{T_{o,ds}}{T_{amb}} \right] \quad (16)$	[35,38,39]
Improvement potential	$IP = (1 - \eta_{Ex}) Ex_{loss} \quad (17)$	[35,39]
<i>Drying system component's performance</i>		
Collector efficiency	$\eta_{sc} = \frac{Q_{EPsc}}{Q_{isc}} \times 100\% \quad (18)$	[14,40]

(continued on next page)

Table 3 (continued)

Parameter	Formula Eq.no.	Ref.
Coefficient of heat pump's performance		[14,19]
	$\text{COP}_{\text{hp}} = \frac{Q_{\text{EPchp}}}{W_{\text{C}}} \quad (19)$	
Biomass furnace efficiency		[32]
	$\eta_{\text{BF}} = \frac{Q_{\text{EPbf}}}{Q_{\text{bfmf}}} \times 100\% \quad (20)$	
Thermal energy input of solar collector		[14,37,40]
	$Q_{\text{isc}} = I_{\text{T}} A_{\text{sc}} \quad (21)$	
Power input of compressor		[14,26]
	$W_{\text{C}} = VI \cos \varphi \quad (22)$	
Power input of dryer blower		[26]
	$W_{\text{Bfd}} = \sqrt{3} VI \cos \varphi \quad (23)$	
Thermal energy input of biomass furnace		[32]
	$Q_{\text{bfmf}} = \dot{m}_{\text{bf}} C_{\text{Pda}} \quad (24)$	
Thermal energy produced by solar collector		[40,41]
	$Q_{\text{EPsc}} = \dot{m}_{\text{a}} C_{\text{a}} (T_{\text{o,sc}} - T_{\text{i,sc}}) \quad (25)$	
Thermal energy produced by condenser of heat pump		[14,19]
	$Q_{\text{EPchp}} = \dot{m}_{\text{a}} C_{\text{a}} (T_{\text{o,chnp}} - T_{\text{i,chnp}}) \quad (26)$	
Thermal energy produced by biomass furnace		[32]
	$Q_{\text{EPbf}} = \dot{m}_{\text{a}} C_{\text{a}} (T_{\text{o,bf}} - T_{\text{i,bf}}) \quad (27)$	
Quality of drying products		
Head rice yield		[42]
	$\text{HRY} = \frac{W_{\text{THR}}}{W_{\text{TS}}} \times 100\% \quad (28)$	
Broken rice		[43]
	$\text{BR} = \frac{W_{\text{TBR}}}{W_{\text{TS}}} \times 100\% \quad (29)$	
Grain grouts		[43]
	$\text{GG} = \frac{W_{\text{TGG}}}{W_{\text{TS}}} \times 100\% \quad (30)$	

Fig. 9 shows the percentages of head rice yield (HRY) obtained after milling the paddy samples dried by the HS-AHPD (present study) compared to those by several drying methods reported in the literature. The percentage of head rice yield of milled sample dried by the HS-AHPD was about $93.10 \pm 1.044\%$. In the dryer, the paddy was dried at air temperatures ranging from 60.0 to 64.8 °C. Xangsayasane et al. [45] found the percentage of head rice yield of paddy was $45.3 \pm 0.17\%$ when the sample was dried by flatbed dryer (FLBD) at an air temperature of 38 – 41 °C. Ibrahim et al. [46] also found the percentage of head rice yield of paddy was $48.9 \pm 0.17\%$ when the drying was conducted by fluidized bed dryer (FBD) at an air temperature of 100 – 120 °C. Meanwhile, Sahari et al. [47] reported that the yield percentage of milled rice reached $73.92 \pm 3.23\%$ when the drying was carried out by inclined bed dryer (IBD) at an air temperature of 45 – 50 °C. The results showed that the head rice yield of paddy was higher when dried with the HS-AHPD (present study) than that with the IBD, FBD, and FLBD. This statement indicates that there was less grain damage while the paddy was dried by the HS-AHPD compared to that dried by the IBD, FBD, and FLBD. This might be due to the uniform paddy moisture content and low drying air temperature in the drying section. A summary of experimental results of drying paddy using HS-AHPD is shown in Table 5.

Table 4

The instruments' characteristics and the parameters' uncertainties during the paddy drying experiment.

Instrument	Function	Range	Accuracy	Uncertainty
Thermocouple (Type: T, Japan)	Measuring air and paddy temperatures	-200 to 400 °C	± 0.1 °C	± 0.20 °C
Pyranometer (Type: LI-200, Japan)	Measuring solar radiation	0 – 2000 Wm^{-2}	± 0.1 Wm^{-2}	± 0.14 W/m^2
Data Logger (Type: AH4000, Japan)	Recording air temperature and solar radiation	–	± 0.1 °C	–
Anemometer (Type: HT-383, China)	Measuring air velocity	0 – 30 ms^{-1}	± 0.2 ms^{-1}	± 0.24 m/s
Grain moisture tester (Type: OEM-MC-7828, China)	Measuring moisture content	0 – 50%	$\pm 0.5\%$	$\pm 0.51\%$ wet basis
Weighing scale (Type: Camry, China)	Measuring the weight of paddy	0 – 100 kg	± 0.1 kg	± 0.14 kg
Weighing scale (Type: TKB, China)	Measuring the weight of biomass fuel	0 – 15 kg	± 0.05 kg	± 0.11 kg

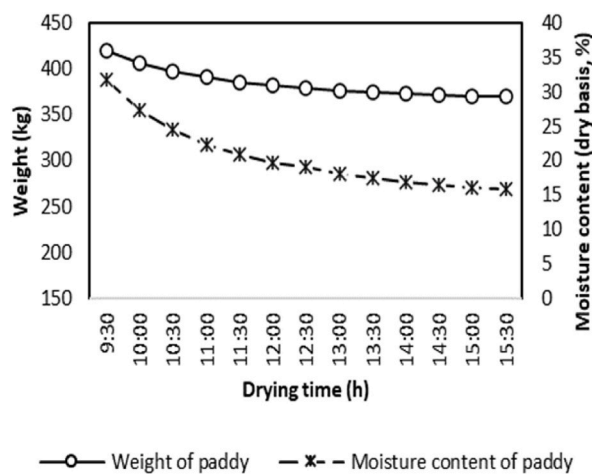


Fig. 2. Variations in weight and MC with drying time.

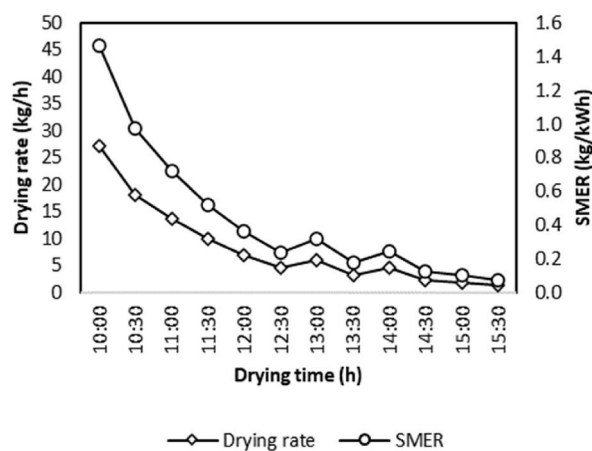


Fig. 3. Variations in DR and SMER with drying time.

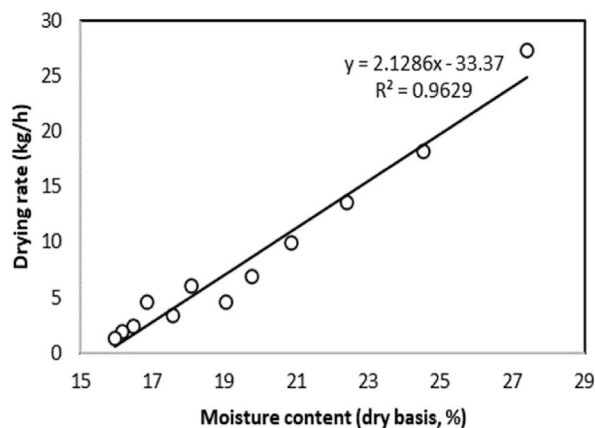


Fig. 4. Variations in DR with MC.

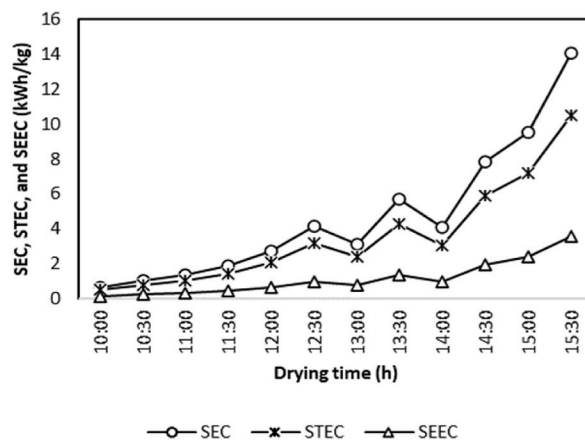


Fig. 5. Variations in SEC, STEC, and SEEC with drying time.

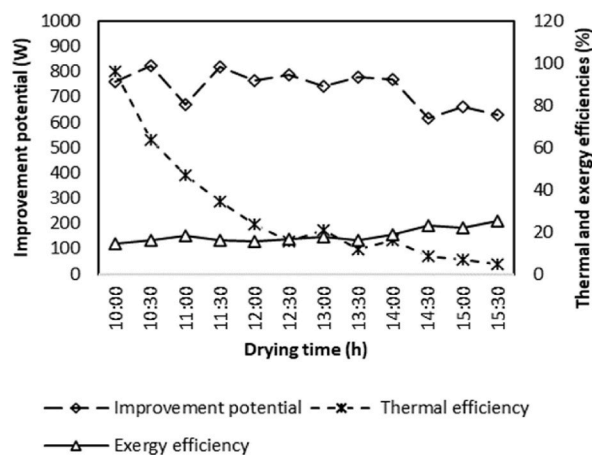


Fig. 6. Variations in IP, thermal, and exergy efficiencies with drying time.

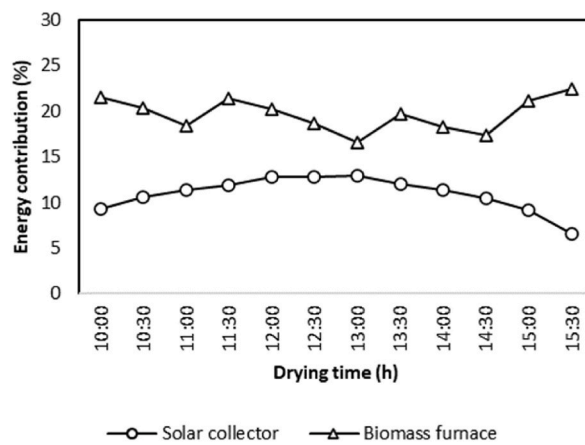


Fig. 7. Variation in energy contribution versus drying time.

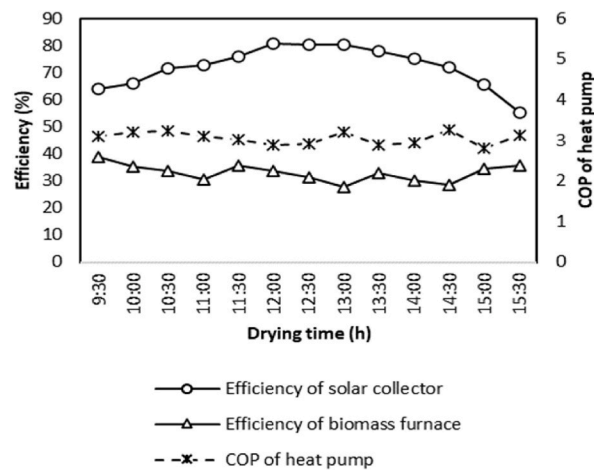


Fig. 8. Variations in efficiency and COP of heat pump versus drying time.

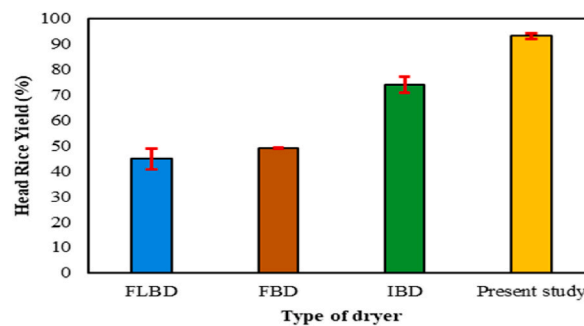


Fig. 9. Head rice yield percentages obtained after milling of the paddy samples desiccated by HS-AHPD compared with several drying methods.

Table 5

Evaluation of the performance of HS-AHPD and the quality of drying products.

Parameter	Unit	Range value	Average value
Drying capacity	kg	–	420
Initial moisture content of paddy (wb)	%	–	24.1
Final moisture content of paddy (wb)	%	–	13.9
Drying air temperature	°C	60.0–64.8	62.9
Drying air relative humidity	%	13.0–19.0	16.1
Drying time	h	–	5.5
SMER	kg/kWh	0.71–1.47	0.44
SEC	kW h/kg	0.68–14.08	4.69
STEC	kW h/kg	0.51–10.48	3.53
SEEC	kW h/kg	0.17–3.59	1.17
Thermal efficiency of HS-AHPD	%	4.7–96.0	29.1
Exergy efficiency	%	14.4–25.3	18.4
Improvement potential	W	628.8–824.4	735.6
COP of heat pump	–	2.6–3.3	3.1
Solar collector efficiency	%	55.4–81.0	72.2
Biomass furnace efficiency	%	27.9–39.0	33.0
Energy contributed by solar collector	%	6.6–12.9	12.9
Energy contributed by biomass furnace	%	17.3–22.4	19.7
Percentage of head rice yield	%	–	93.10 ± 1.044
Percentage of broken rice	%	–	4.41 ± 0.737
Percentage of grain grouts	%	–	1.11 ± 0.271

4. Conclusion

This study was aimed at designing, constructing, and evaluating an HS-AHPD. The drying experiment results showed that.

- During the drying experiment, in general, the thermal energy used to heat the desiccating air were contributed by biomass fuel and solar energy.
- The average efficiency value of the solar collector was over 72.2%, while that of the biomass furnace was over 33.0% and the average COP of the heat pump was 3.1.
- In the HS-AHPD, reducing the paddy's weight from 420 kg with an MC of 31.67% db (24.1% wb) to 370.60 kg with an MC of 16.18% db (13.9% wb) needed approximately 5.5 h with an average air temperature of 62.9 °C and an average relative humidity of 16.1%, at a mass flow rate of 0.195 kg/s.
- The average DR and SMER were 8.34 kg/s and 0.44 kg/kWh, respectively.
- The average SEC, STEC, and SEEC were 4.694, 3.53, and 1.17 kWh/kg, respectively.
- The drying system's thermal efficiency the drying section's exergy efficiency were found to be over 29.1% and 18.4%, respectively, and the average IP was 735.6 W.
- The furnace and solar collector contributed energy over 19.79% and 12.9%, respectively.
- The drying resulted in higher-quality products: higher head rice yield ($93.10 \pm 1.044\%$), lower broken rice ($4.41 \pm 0.737\%$), and lower rice grouts ($1.11 \pm 0.271\%$).

Author statement

The authors declare that they have no known conflicting financial interests or personal ties that might have influenced the research presented in this study.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

- [1] A. Singh, J. Sarkar, R.R. Sahoo, Experimental energy, exergy, economic and exergoeconomic analyses of batch-type solar-assisted heat pump dryer, *Renew. Energy* 156 (2020) 1107–1116.
- [2] M. Yahya, A. Fudholi, H. Hafizh, K. Sopian, Comparison of solar dryer and solar-assisted heat pump dryer for cassava, *Sol. Energy* 136 (2016) 606–613.
- [3] M. Mohanraj, Ye Belyayev, S. Jayaraj, A. Kaltayev, Research and developments on solar assisted compression heat pump systems-A comprehensive review (Part B: applications), *Renew. Sustain. Energy Rev.* 83 (2018) 124–155.
- [4] C. Strumillo, P.L. Jones, R.Z. Ylla, Energy aspects in drying, in: A.S. Mujumdar (Ed.), *Handbook of Industrial Drying*, CRC Press, Boca Raton, FL, 2006, pp. 1075–1099.
- [5] M. Mohanraj, Ye Belyayev, S. Jayaraj, A. Kaltayev, Research and developments on solar assisted compression heat pump systems - a comprehensive review (Part A: modeling and modifications), *Renew. Sustain. Energy Rev.* 83 (2018) 90–123.
- [6] M. Mohanraj, P. Chandrasekar, V.V. Sreenarayanan, Performance of a heat pump drier for copra drying, *Proc. Inst. Mech. Eng. A J. Power Energy* 222 (3) (2008) 283–287.
- [7] F. Afshari, O. Comakli, N. Adiguzel, S. Karagoz, Optimal charge amount for different refrigerants in air-to-water heat pumps, *Iranian J. Sci. Technol. Trans. Mech. Eng.* 40 (2016) 325–335.
- [8] F. Afshari, A. Sözen, A. Khanlari, A.D. Tuncer, H.M. Ali, Experimental investigation of effect of refrigerant gases, compressor lubricant and operating conditions on performance of a heat pumps, *J. Cent. S. Univ.* 28 (11) (2021) 3556–3568.
- [9] R. Best, W. Soto, I. Pilatowsky, L.J. Gutierrez, Evaluation of a rice drying system using a solar assisted heat pump, *Renew. Energy* 5 (1994) 465–468.
- [10] H. Li, Y. Dai, J. Dai, X. Wang, L. Wei, A solar assisted heat pump drying system for grain in-store drying, *Front. Energy Power Eng.* 4 (3) (2010) 386–391.
- [11] M.N.A. Hawlader, K.A. Jahangeer, Solar heat pump drying and water heating in the tropics, *Sol. Energy* 80 (2006) 492–499.
- [12] M.M.S. Dezfouli, S. Mat, M.H. Ruslan, K. Sopian, Evaluation of drying chili by two methods: solar assisted heat pump dryer and open sun drying, in: *Proceeding Of the 1st International Conference On Environmental Informatics (ENINF'13)*, Kuala Lumpur, Malaysia, 2013, pp. 112–116.
- [13] S.M.A. Rahman, R. Saidur, M.N.A. Hawlader, An economic optimization of evaporator and air collector area in a solar assisted heat pump drying system, *Energy Convers. Manag.* 76 (2013) 377–384.
- [14] S. Sevik, Experimental investigation of a new design solar-heat pump dryer under the different climatic conditions and drying behavior of selected products, *Sol. Energy* 105 (2014) 190–205.
- [15] Y. Qiu, M. Li, R.H.E. Hassanien, Y. Wang, X. Luo, Q. Yu, Performance and operation mode analysis of a heat recovery and thermal storage solar-assisted heat pump drying system, *Sol. Energy* 137 (2016) 225–235.
- [16] Y. Wang, M. Li, Y. Qiu, Q. Yu, X. Luo, G. Li, X. Ma, Performance analysis of a secondary heat recovery solar-assisted heat pump drying system for mango, *Energy Explor. Exploit.* 37 (4) (2019) 1377–1387.
- [17] H.H. Ismaeel, R. Yumrutas, Investigation of a solar assisted heat pump wheat drying system with underground thermal energy storage tank, *Sol. Energy* 199 (2020) 538–551.

- [18] J. Li, Y. Zhang, M. Li, Y. Wang, M. Shi, M. Gao, Z. Deng, G. Lu, R. Liu, Study on heating performance of solar-assisted heat pump drying system under large temperature difference, *Sol. Energy* 229 (2021) 148–161.
- [19] S. Sevik, M. Aktas, H. Dogan, S. Kocak, Mushroom drying with solar assisted heat pump system, *Energy Convers. Manag.* 72 (2013) 171–178.
- [20] M. Mohanraj, Performance of a Solar-Ambient Hybrid Source Heat Pump Drier for Copra Drying under Hot-Humid Weather Conditions, vol. 23, *Energy for Sustainable Development*, 2014, pp. 165–169.
- [21] I. Ceylan, A.E. Gürel, Solar-assisted fluidized bed dryer integrated with a heat pump for mint leaves, *Appl. Therm. Eng.* 106 (2016) 899–905.
- [22] M. Kuan, Ye Shaker, M. Mohanraj, Ye Belyayev, S. Jayaraj, A. Kaltayev, Numerical simulation of a heat pump assisted solar dryer for continental climates, *Renew. Energy* 143 (2019) 214–225.
- [23] T. Naemsai, J. Jareanjit, K. Thongkaew, Experimental investigation of solar-assisted heat pump dryer with heat recovery for the drying of chili peppers, *J. Food Process. Eng.* (2019) 1–10.
- [24] X. Gu, J. Dai, H. Li, Y. Dai, Experimental and theoretical assessment of a solar-assisted heat pump system for in-bin grain drying: a comprehensive case study, *Renew. Energy* 181 (2022) 426–444.
- [25] M. Yahya, Design and performance evaluation of a solar assisted heat pump dryer integrated with biomass furnace for red chilli, *Int. J. Photoenergy* (2016) 1–14.
- [26] M. Yahya, H. Fahmi, A. Fudholi, K. Sopian, Performance and economic analyses on solar-assisted heat pump fluidized bed dryer integrated with biomass furnace for rice drying, *Sol. Energy* 174 (2018) 1058–1067.
- [27] BPS (Badan Pusat Statistik Indonesia), Statistik Indonesia, Jakarta, 2022.
- [28] B. Prastowo, Biomass Resource in Indonesia: Indonesia's Solid Biomass Energy Potential," Presented in Indonesia-German Workshop and Seminar, Sep. 26, Institute Technology of Bandung, 2011, pp. 1–15.
- [29] N.A. Handayani, D. Ariyanti, Potency of solar energy application in Indonesia, *Int. J. Renew. Energy Dev.* 1 (2) (2012) 33–38.
- [30] H.K. Purwadaria, Problems and priorities of grain drying in Indonesia, in: *Grain Drying in Asia. Proceedings of an International Conference Held at the 74 FAO Regional Office for Asia and the Pacific*, 1995, pp. 201–209. Bangkok, Thailand.
- [31] BSN (Badan Standardisasi Nasional), Standar Nasional Indonesia Beras Giling, SNI 6128: 2008, vol. 9, Badan Standardisasi Nasional, Jakarta, 2008.
- [32] M. Yahya, H. Fahmi, H. Hasibuan, Experimental performance analysis of a pilot-scale biomass-assisted recirculating mixed-flow dryer for drying paddy, *Int. J. Food Sci.* (2022) 1–15.
- [33] M. Yahya, Performance analysis of solar assisted fluidized bed dryer integrated biomass furnace with and without heat pump for drying of paddy, *Int. J. Photoenergy* (2016) 1–17.
- [34] E.K. Akpinar, Drying of mint leaves in solar dryer and under open sun: modelling, performance analyses, *Energy Convers. Manag.* 51 (2010) 2407–2418.
- [35] M. Yahya, A. Rachman, R. Hasibuan, Performance analysis of solar-biomass hybrid heat pump batch-type horizontal fluidized bed dryer using multi-stage heat exchanger for paddy drying, *Energy* 254 (2022), 124294.
- [36] S. Mehran, M. Nikian, M. Ghazi, H. Zareiforush, I. Bagheri, Experimental investigation and energy analysis of a solar-assisted fluidized bed dryer including solar water heater and solar-powered infrared lamp for paddy grains drying, *Sol. Energy* 190 (2019) 167–184.
- [37] A. Fudholi, K. Sopian, M.Y. Yazdi, M.H. Ruslan, M. Gabbasa, H.A. Kazem, Performance analysis of solar drying system for red chili, *Sol. Energy* 99 (2014) 47–54.
- [38] M.S.N. Sarker, M.N. Ibrahim, A.B. Aziz N, M.S. Punan, Energy and exergy analysis of industrial fluidized bed drying of paddy, *Energy* 84 (1) (2015) 131–138.
- [39] A. Fudholi, K. Sopian, M.Y. Othman, M.H. Ruslan, Energy and exergy analyses of solar drying system of red seaweed, *Energy Build.* 68 (2014) 121–129.
- [40] J. Banout, P. Ehl, J. Havlik, B. Lojka, Z. Polesny, V. Verner, Design and performance evaluation of a Double-pass solar drier for drying of red chilli (*Capsicum annum* L.), *Sol. Energy* 85 (3) (2011) 506–515.
- [41] A. Fudholi, K. Sopian, M.A. Alghoul, M.H. Ruslan, M.Y. Othman, Performances and improvement potential of solar drying system for palm oil fronds, *Renew. Energy* 78 (2015) 561–565.
- [42] M.A. Wazed, N.H.M.R. Mozumder, M.S.H. Sarker, Effect of two stage drying employing fluidized bed, tempering followed by fixed bed drying on head rice yield of BRRIDHAN 28 rice variety in Bangladesh, *Sustain. Food. Agri.(SFNA)* 2 (2) (2021) 74–78.
- [43] M.R. Alizadeh, Effect of paddy husked ratio on rice breakage and whiteness during milling process, *Aust. J. Crop. Sci.* 5 (5) (2011) 562–565.
- [44] Badan Standardisasi Nasional, *Standar Nasional Indonesia* 6128: 2015, Beras, Jakarta, 2015.
- [45] P. Xangsayasane, K. Vongxayya, S. Phongchanmisai, J. Mitchell, S. Fukai, Rice milling quality as affected by drying method and harvesting time during ripening in wet and dry seasons, *Plant Prod. Sci.* (2018) 1–10.
- [46] M.N. Ibrahim, M.S.H. Sarker, A.B. Aziz N, M. Salleh, Drying performances and milling quality of rice during industrial fluidized bed drying of paddy in Malaysia, *Pertanika J. Sci & Technol.* 23 (2) (2015) 297–309.
- [47] Y. Sahari, R.A. Wahid, A.S.M. Adnan, M. Sairi, H. Hosni, E.H.E. Abdullah, S. Alwi, M.H.M.A. Tawakkal, M.Z.Z. Abidin, Z. Aris, Study on the drying performance and milling quality of dried paddy using inclined bed dryers in two different paddy mills located in MADA and IADA KETARA, *Int. Food Res. J.* 25 (6) (2018) 2572–2578.